

Research Report

Query: mathematical proof of gradient descent convergence

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Executive Summary

- Gradient descent is a widely used optimization algorithm that converges to a local minimum of a function under certain conditions. The convergence is guaranteed for convex functions with Lipschitz continuous gradients when the step size is appropriately chosen.

Key Points

- Gradient descent iteratively updates parameters in the direction of the negative gradient.
- Convergence is assured for convex functions with Lipschitz continuous gradients when the step size is less than or equal to $1/L$, where L is the Lipschitz constant.

Important Findings

- For convex and differentiable functions, the convergence rate is $O(1/k)$, meaning the error decreases inversely with the number of iterations.
- The choice of step size is critical; too large a step size can lead to divergence.

Actionable Insights

- Implement backtracking line search to adaptively choose the step size for better convergence.
- Ensure that the function being minimized is convex and that its gradient is Lipschitz continuous to guarantee convergence.

References

- Gradient Descent: Convergence Analysis <https://www.stat.cmu.edu/~ryantibs/convexopt-F13/scribes/lec6.pdf>
- Convergence Rate of Gradient Descent <https://jhc.sjtu.edu.cn/~kuanyang/teaching/MATH3806/notes/lec10.pdf>
- Gradient descent and its convergence analysis https://mmids-textbook.github.io/chap03_opt/05_gd/roch-mmids-opt-gd.html
- Convergence of Gradient Descent <https://www.cs.ubc.ca/~schmidtm/Courses/540-W18/L4.pdf>
- Gradient Descent - Wikipedia https://en.wikipedia.org/wiki/Gradient_descent